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Research article

Limited transgenerational effects of environmental temperatures on thermal performance of a cold-adapted salmonid

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The capacity of ectotherms to cope with rising temperatures associated with climate change is a significant conservation concern as the rate of warming is likely too rapid to allow for adaptative responses in many populations. Transgenerational plasticity (TGP), if present, could potentially buffer some of the negative impacts of warming on future generations. We examined TGP in lake trout to assess their inter-generational potential to cope with anticipated warming. We acclimated adult lake trout to cold (10°C) or warm (17°C) temperatures for several months, then bred them to produce offspring from parents within a temperature treatment (cold-acclimated and warm-acclimated parents) and between temperature treatments (i.e. reciprocal crosses). At the fry stage, offspring were also acclimated to cold (11°C) or warm (15°C) temperatures. Thermal performance was assessed by measuring their critical thermal maximum (CTM) and the change in metabolic rate during an acute temperature challenge. From this dataset, we also determined their resting and peak (highest achieved, thermally induced) metabolic rates. There was little variation in offspring CTM or peak metabolic rate, although cold-acclimated offspring from warm-acclimated parents exhibited elevated resting metabolic rates without a corresponding increase in mass or condition factor, suggesting that transgenerational effects can be detrimental when parent and offspring environments mismatch. These results suggest that the limited TGP in thermal performance of lake trout is unlikely to significantly influence population responses to projected increases in environmental temperatures.

Key words: Climate change, critical thermal maximum, transgenerational plasticity, lake trout, thermal tolerance, epigenetic

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Introduction

Populations are being forced to respond to climate change as environmental temperatures continue to increase towards their viable limits (Hazen *et al.*, 2013; Galbraith *et al.*, 2014; Luo *et al.*, 2015). Many species are resorting to migration and range shifts where movement to more suitable habitats is possible (e.g. freshwater fish, Chu *et al.*, 2005; birds, VanDerWal *et al.*, 2013; and mussels, Inoue & Berg, 2017), but those that are unable to relocate will need to acclimatize